



Calculation Sheet

Calculation header

Identifier

*New calculation (1)***Medium selection and state**

Medium

Water

State

Vaporous

Critical pressure

pc

220.64

bar(a)

Operating data☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	<i>280.0</i>	<i>280.0</i>	<i>280.0</i>	°C
p1	<i>18.0</i>	<i>9.0</i>	<i>0.05</i>	kgf/cm ² (g)
☐ p2	<i>0.0</i>	<i>2.0</i>	<i>0.0</i>	kgf/cm ² (g)
☒ Δp	<i>18.0</i>	<i>7.0</i>	<i>0.05</i>	kgf/cm ²
Cv	<i>1.0403</i>	<i>0.15675</i>		GPM(US)
☒ qm	<i>198.0</i>	<i>16.0</i>	☒ <i>0.0</i>	kg/h
☐ qv	<i>25.586</i>	<i>4.0349</i>		m ³ /h
s/s100	<i>100.0</i>	<i>58.334</i>		%
LpAe	<i>66.0</i>	<i>40.1</i>		dB(A)
	<i>Choked flow!</i>	<i>Non-critical</i>	<i>Non-critical</i>	

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ρ1	<i>7.7385</i>	<i>3.9654</i>	<i>0.41733</i>	kg/m ³
ρ2	<i>0.45767</i>	<i>1.1967</i>	<i>0.39806</i>	kg/m ³
☒ η1	<i>0.01921</i>	<i>0.01933</i>	<i>0.01946</i>	mPa s
☐ ν1	<i>2.4824</i>	<i>4.8747</i>	<i>46.63</i>	mm ² /s
pv1	<i>64.166</i>	<i>64.166</i>	<i>64.166</i>	bar(a)
pv2	<i>18.657</i>	<i>55.244</i>	<i>64.101</i>	bar(a)
tv1	<i>208.91</i>	<i>179.17</i>	<i>101.3</i>	°C
cF1	<i>558.84</i>	<i>567.68</i>	<i>575.71</i>	m/s
κ	<i>1.2948</i>	<i>1.2988</i>	<i>1.3021</i>	-
Z1	<i>0.94481</i>	<i>0.97195</i>	<i>0.99708</i>	-
Z2	<i>0.99517</i>	<i>0.99115</i>	<i>0.99721</i>	-





Calculation Sheet

Pipe downstream of valve

Size class downstream of valve	NPS2	3/4"
Schedule downstream	SCH2	STD

Valve configuration

Valve type	<i>Straight globe valve</i>
Trim type	<i>Contoured plug</i>
Flow direction	<i>FTO</i>
Valve performance class	<i>Heavy duty valve</i>
Protection	<i>Hardened seat/plug</i>
Low-noise design	<i>n.a.</i>

Valve data

Basic characteristic		<i>Equal percentage</i>	
Nominal flow coefficient	Cv100	<i>0.8</i>	GPM(US)
☐ Lift stopper			
Selected valve size	NPS	<i>3/4"</i>	
Pressure class	class	<i>class 600</i>	
Maximum operating pressure	p,max	<i>105.0</i>	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL ²	<i>0.928</i>	-
Valve modifier (Cv/Cv100 = 0.75)	xT	<i>0.798</i>	-
Valve style modifier (Cv/Cv100 = 1)	Fd	<i>0.41</i>	-

Load-dependent values

	Maximum flow	Mean flow	Minimum flow	
s/s100	<i>100.0</i>	<i>58.334</i>		%
u2	<i>421.63</i>	<i>13.03</i>		m/s
MaDN	<i>0.75447</i>	<i>0.022954</i>		-
LpAe	<i>66.0</i>	<i>40.1</i>		dB(A)
P	<i>28.251</i>	<i>1.1533</i>		kW
Ri	<i>0.39</i>	<i>0.0</i>		-





Calculation Sheet

Noise calculation

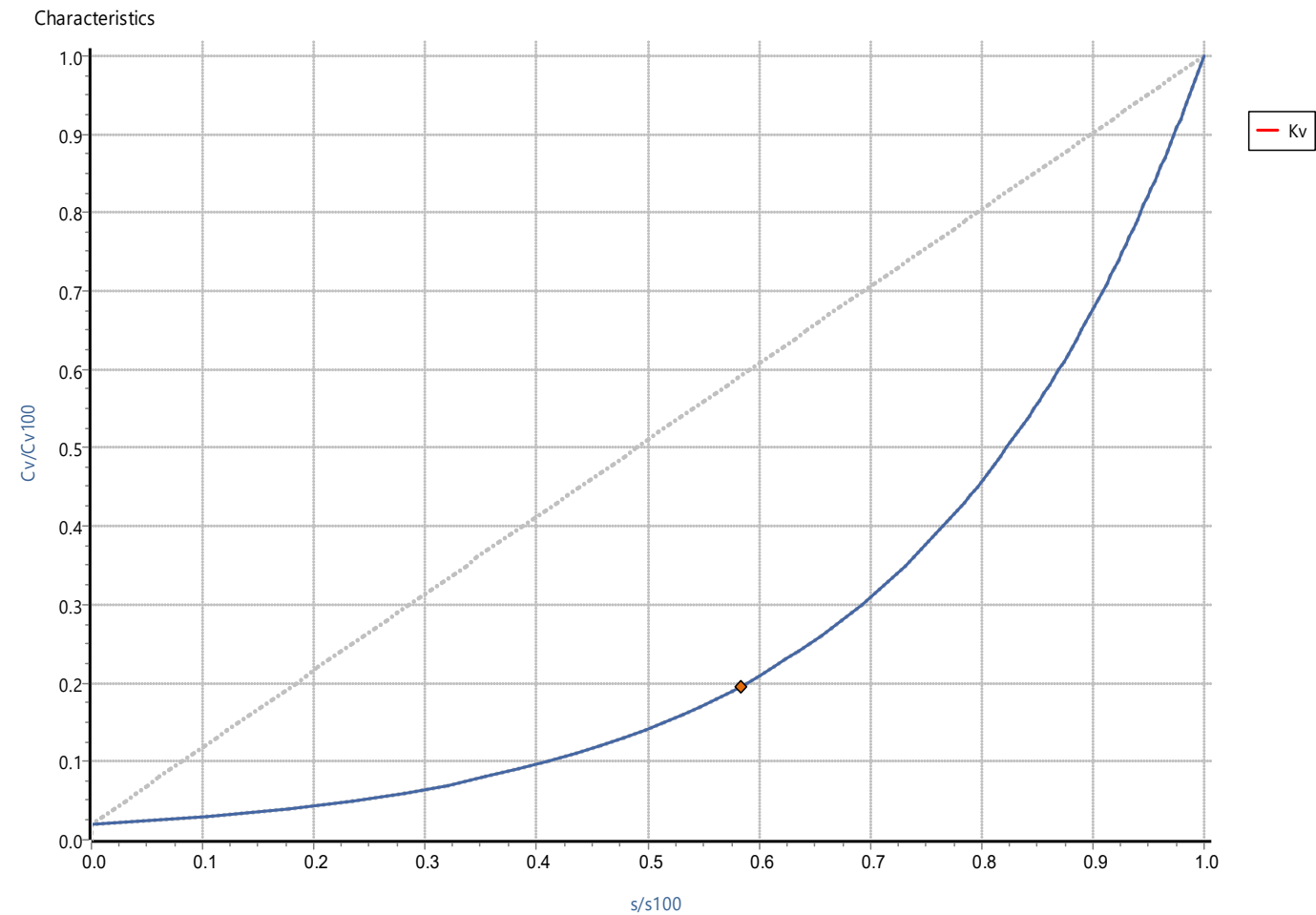
Calculation standard (noise, gas) IEC 60534-8-3 (2010)

Noise prediction data

Low-noise design n.a.
Sound pressure level of valve (A-weighted) LpAe 66.0 dB(A)

Minor noise prediction data

Mach number (valve outlet) MaDN 0.75447 -
Power loss P 28.251 kW





Calculation Sheet

Comments:

Reliability index - Ri

*High velocity: 31 %; Outlet/piping: 69 %
Recommended solution: Heavy duty valve*

Reliability index - Ri,op2

Recommended solution: Standard valve

Legend

 Errors

